Maths Assignment 2

Q1a) $\lim_{x \to 0} \left(\frac{1}{x\sin x} - \frac{1}{x\tan x} \right)$

Sol

$= \lim_{x \to 0} \left(\frac{1}{x\sin x} - \frac{\cos x}{x\sin x} \right)$

$= \lim_{x \to 0} \left(\frac{1-\cos x}{x\sin x} \right)$ → indeterminat form $\frac{0}{0}$

applying L'Hopital Rule

derivative of $x\sin x = vu' + uv'$

$= \sin x(1) + x(\cos x)$

$= \sin x + x\cos x$

derivative of $1-\cos x \Rightarrow 0 + \sin x$

$= \sin x$

$= \lim_{x \to 0} \left(\frac{\sin x}{\sin x + x\cos x} \right)$ → again indeterminate form $\frac{0}{0}$

applying L'Hopital Rule

derivative of $\sin x + x\cos x$

$\Rightarrow \cos x + \cos x(1) + x(-\sin x)$

$2\cos x - x\sin x$

derivative of $\sin x \Rightarrow \cos x$

$= \lim_{x \to 0} \left(\frac{\cos x}{2\cos x - x\sin x} \right)$

$= \frac{\cos(0)}{2\cos(0) - 0\sin(0)}$

$\boxed{= \frac{1}{2}}$ Answer

Q1b) $\lim_{x \to -\infty} \left(x - \sqrt{x^2 + 3x} \right)$

Sol

$$\lim_{x \to -\infty} \left[\left(x - \sqrt{x^2+3x}\right) \times \frac{\left(x + \sqrt{x^2+3x}\right)}{\left(x + \sqrt{x^2+3x}\right)} \right]$$

$$\lim_{x \to -\infty} \left[\frac{x^2 + x\sqrt{x^2+3x} - x\sqrt{x^2+3x} - x^2 + 3x}{x + \sqrt{x^2+3x}} \right]$$

$$\lim_{x \to -\infty} \left[\frac{3x}{x + \sqrt{x^2\left(1+\frac{3}{x}\right)}} \right]$$

$$\lim_{x \to -\infty} \left[\frac{3x}{x\left(1 + \sqrt{1+\frac{3}{x}}\right)} \right]$$

$$\lim_{x \to -\infty} \left[\frac{3}{1 + \sqrt{1+\frac{3}{x}}} \right]$$

$$= \frac{3}{1 + \sqrt{1 + \frac{3}{-\infty}}}$$

$$= \frac{3}{1 + \sqrt{1}}$$

$$\boxed{= \frac{3}{2}}$$ Answer

Q1c) $\lim_{x \to +\infty} \left(x - \ln(x^2+3)\right)$

sol

$$\lim_{x \to +\infty} \left(\ln e^x - \ln(x^2+3)\right)$$

$$\lim_{x \to +\infty} \left(\ln\left(\frac{e^x}{x^2+3}\right)\right)$$ $\Rightarrow$ indeterminant form $\frac{\infty}{\infty}$

applying L Hopital rule

derivative of $e^x \Rightarrow e^x.(1)$
$= e^x$

derivative of $x^2+3 \Rightarrow 2x+0$
$= 2x$

$$\lim_{x \to +\infty} \left(\ln\left(\frac{e^x}{2x}\right)\right)$$ $\rightarrow$ again indeterminant form

applying L Hopital Rule

derivative of $e^x \Rightarrow e^x.(1)$

derivative of $2x \Rightarrow 2$

$$\lim_{x \to +\infty} \left(\ln\left(\frac{e^x}{2}\right)\right)$$

$$\Rightarrow \ln\left(\frac{e^\infty}{2}\right)$$

$$\boxed{= +\infty}$$ Answer

Q2) $f(x) = 5x^{\frac{1}{3}} - x^{\frac{5}{3}}$

$f(1) = 4$

$f(-1) = -4$

$f'(x) = 0$

$\frac{5}{3} x^{\frac{2}{3}} - \frac{5}{3} x^{-\frac{2}{3}} = 0$

$\frac{5}{3}\left(x^{\frac{2}{3}} - \frac{1}{x^{\frac{2}{3}}}\right) = 0$

$\frac{x^{\frac{4}{3}} - 1}{x^{\frac{2}{3}}} = 0$

$f'(x)$ is undefined at $x = 0$

$x^{\frac{4}{3}} = 1$

$\left(\sqrt[3]{x^4}\right)^3 = (1)^3$

$\sqrt{x^4} = \sqrt{1}$

$x = \pm 1$

$(-\infty, -1) \cup (-1, 0) \cup (0, 1) \cup (1, \infty)$

$f'(-2) = -ve$
$f'(-0.5) = +ve$ } decreasing - increase $\smile$

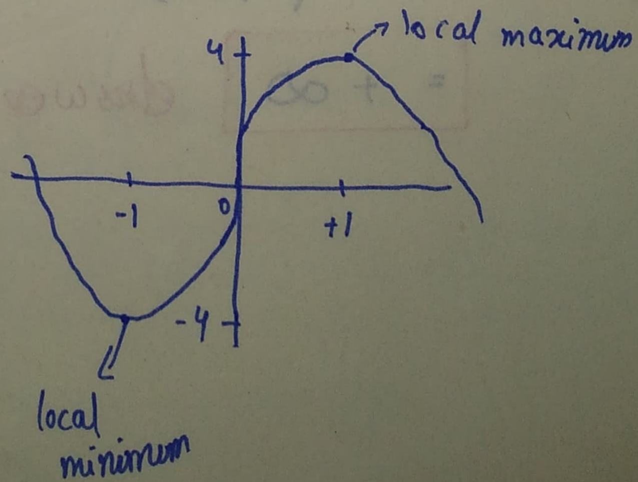
$\therefore$ f has a local minimum at $x = -1$

$f'(-0.5) = +ve$
$f'(0.5) = +ve$

$\therefore$ f has a local maximum at $x = 1$

$f'(0.5) = +ve$
$f'(-0.5) = +ve$

$f'(0.5) = +ve$
$f'(2) = -ve$ } increasing - decreasing $\frown$



Q3) $f(x) = x^3 - 6x^2 + 9x + 30$

$f'(x) = 3x^2 - 12x + 9$ ↗ critical point at $x = 1, 3$

$f(1) = 34$

$f(3) = 30$

$f(2) = 32$

$f''(x) = 6x - 12$

$f''(x) = 0$

$6x - 12 = 0$

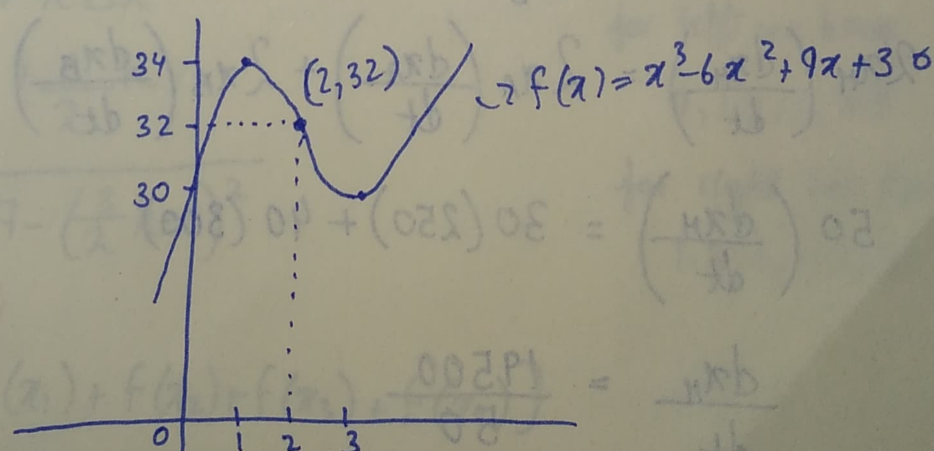
$x = \frac{12}{6}$

$x = 2$

interval → $(-\infty, 2)$ and $(2, \infty)$

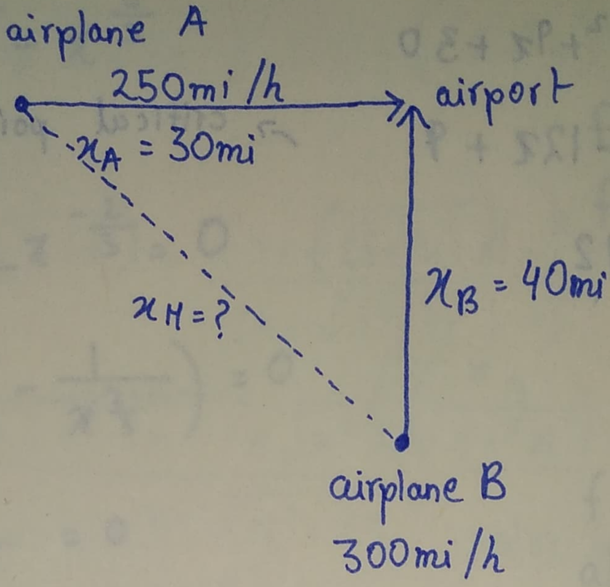
$f''(0) = -ve$ concave down

$f''(3) = +ve$ concave up



The given function has a point of inflection at (2,32) where the graph changes concavity.

Q4)



need to find $\frac{dx}{dt}$ b/w two planes

$$\sqrt{x_H^{\,x}} = \sqrt{x_A^2 + x_B^2}$$

$$x_H = \sqrt{30^2 + 40^2}$$

$$x_H = 50 \text{ miles}$$

$$x_H^2 = x_A^2 + x_B^2$$

$$2x_H\left(\frac{dx_H}{dt}\right) = 2x_A\left(\frac{dx_A}{dt}\right) + 2x_B\left(\frac{dx_B}{dt}\right)$$

$$50\left(\frac{dx_H}{dt}\right) = 30(250) + 40(300)$$

$$\frac{dx_H}{dt} = \frac{19500}{50}$$

$$\boxed{\frac{dx_H}{dt} = 390 \text{ mi/hr}}$$

Answer

Q5) Left end Point $\qquad f(x) = \sqrt{9-(x-3)^2}$

$$\Delta x = \frac{b-a}{n} = \frac{6-0}{4} = \frac{3}{2}$$

$$x_x = a + (k-1)\Delta x$$
$$= 0 + (k-1)\left(\frac{3}{2}\right)$$
$$= \frac{3}{2}(k-1)$$

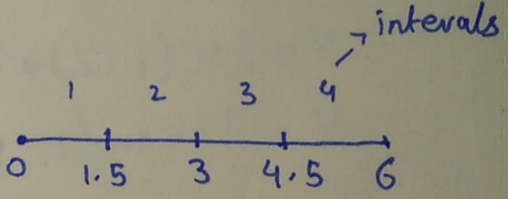
$$f(x_x) = \sqrt{9 - \left(\frac{3(k-1)}{2} - 3\right)^2}$$

Right End Point

$$\Delta x = \frac{b-a}{n} = \frac{6-a}{4} = \frac{3}{2}$$

$$x_x = a + k\Delta x$$
$$x_x = 0 + \frac{3}{2}k$$

$$f(x_x) = \sqrt{9 - \left(\frac{3}{2}k - 3\right)^2}$$



for left we have
0, 1.5, 3, 4.5

for right we have
1.5, 3, 4.5, 6

for left (L_4)

$$A_L = \Delta x\left[f(x_1) + f(x_2) + f(x_3) + f(x_4)\right]$$
$$= \frac{3}{2}\left[f(0) + f(1.5) + f(3) + f(4.5)\right]$$
$$= \frac{3}{2}\left[0 + \frac{3\sqrt{3}}{2} + 3 + \frac{3\sqrt{3}}{2}\right]$$
$$= \frac{9 + 9\sqrt{3}}{2}$$
$$\boxed{\simeq 12.29} \text{ Ans}$$

$$A_R = \Delta x\left[f(x_1)+f(x_2)+f(x_3)+f(x_4)\right]$$ for Right (R_4)

$$= \frac{3}{2}\left[f(1.5)+f(3)+f(4.5)+f(6)\right]$$

$$= \frac{3}{2}\left[\frac{3\sqrt{3}}{2}+3+\frac{3\sqrt{3}}{2}+0\right]$$

$$= \frac{9+9\sqrt{3}}{2}$$

$$\boxed{\simeq 12.29}$$ Ans

as $L_4 = R_4$ we have solve it correctly.